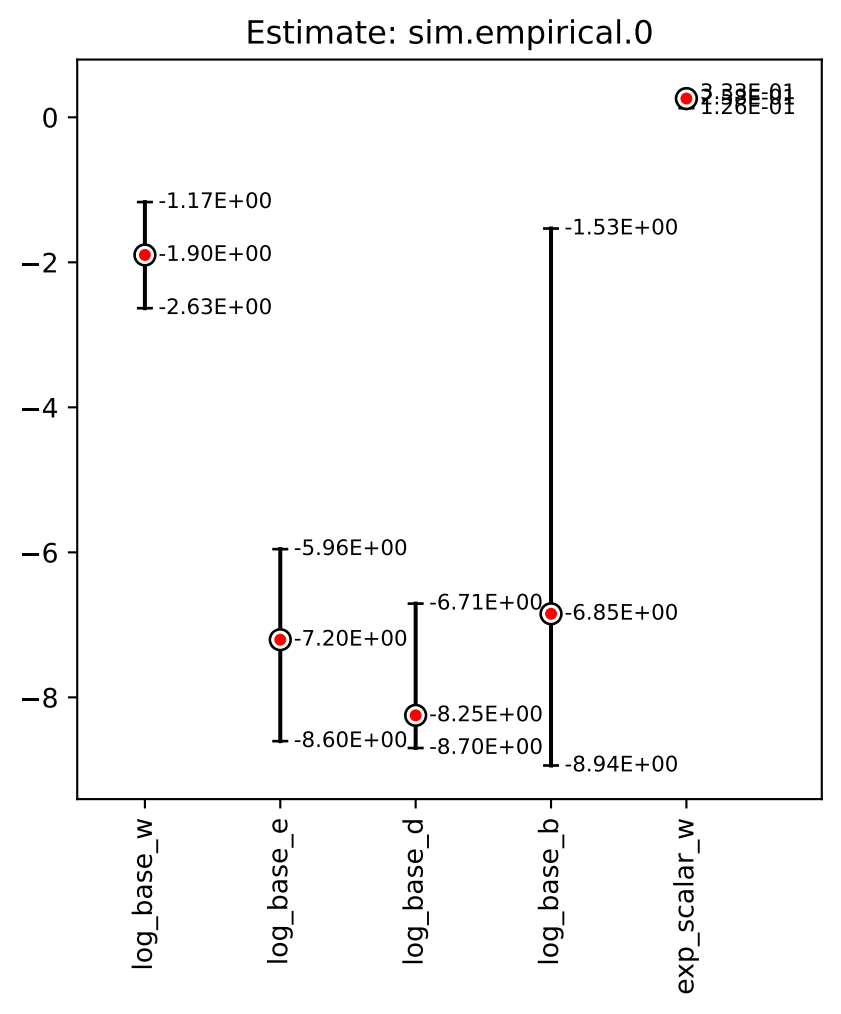
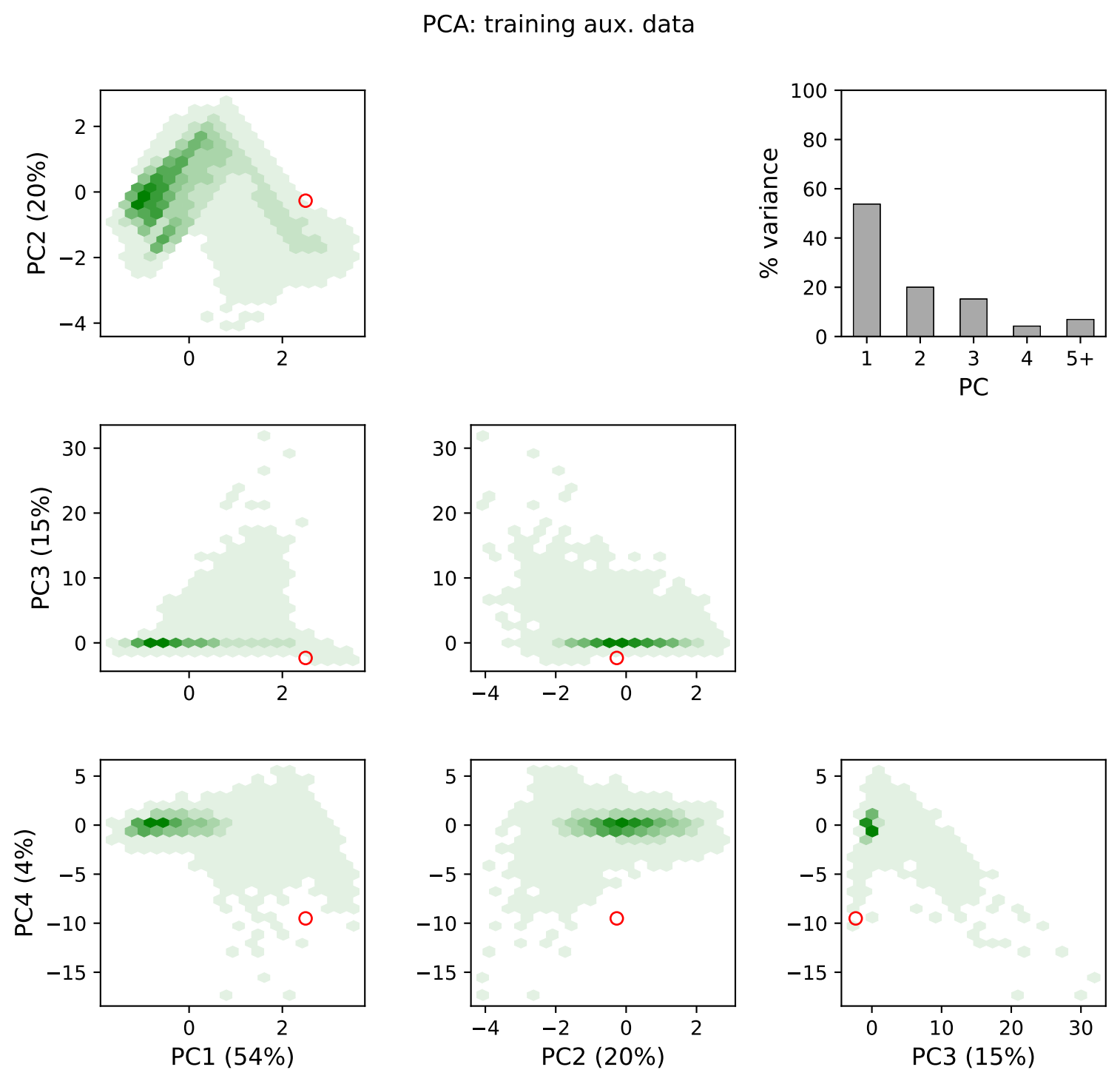
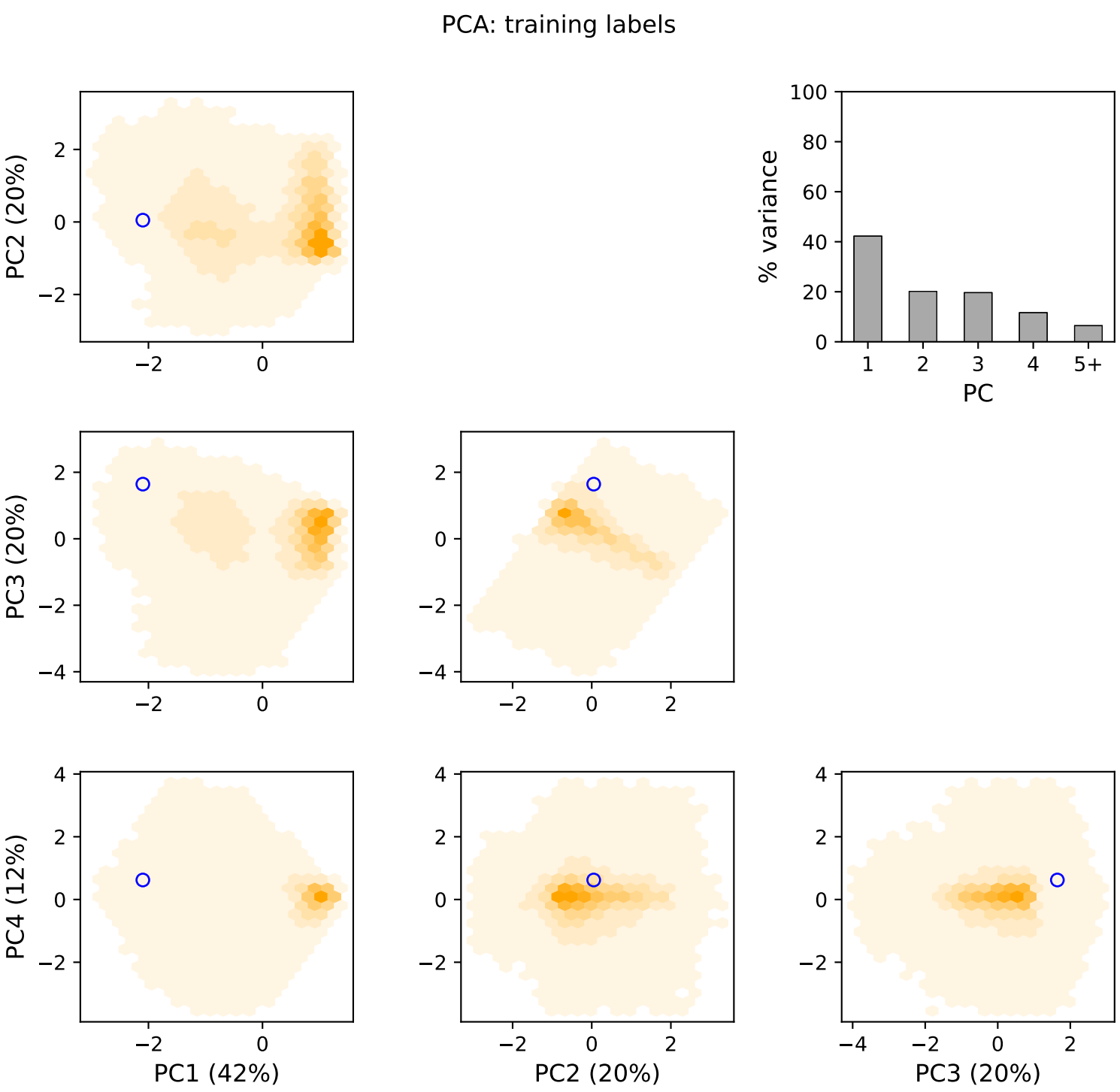
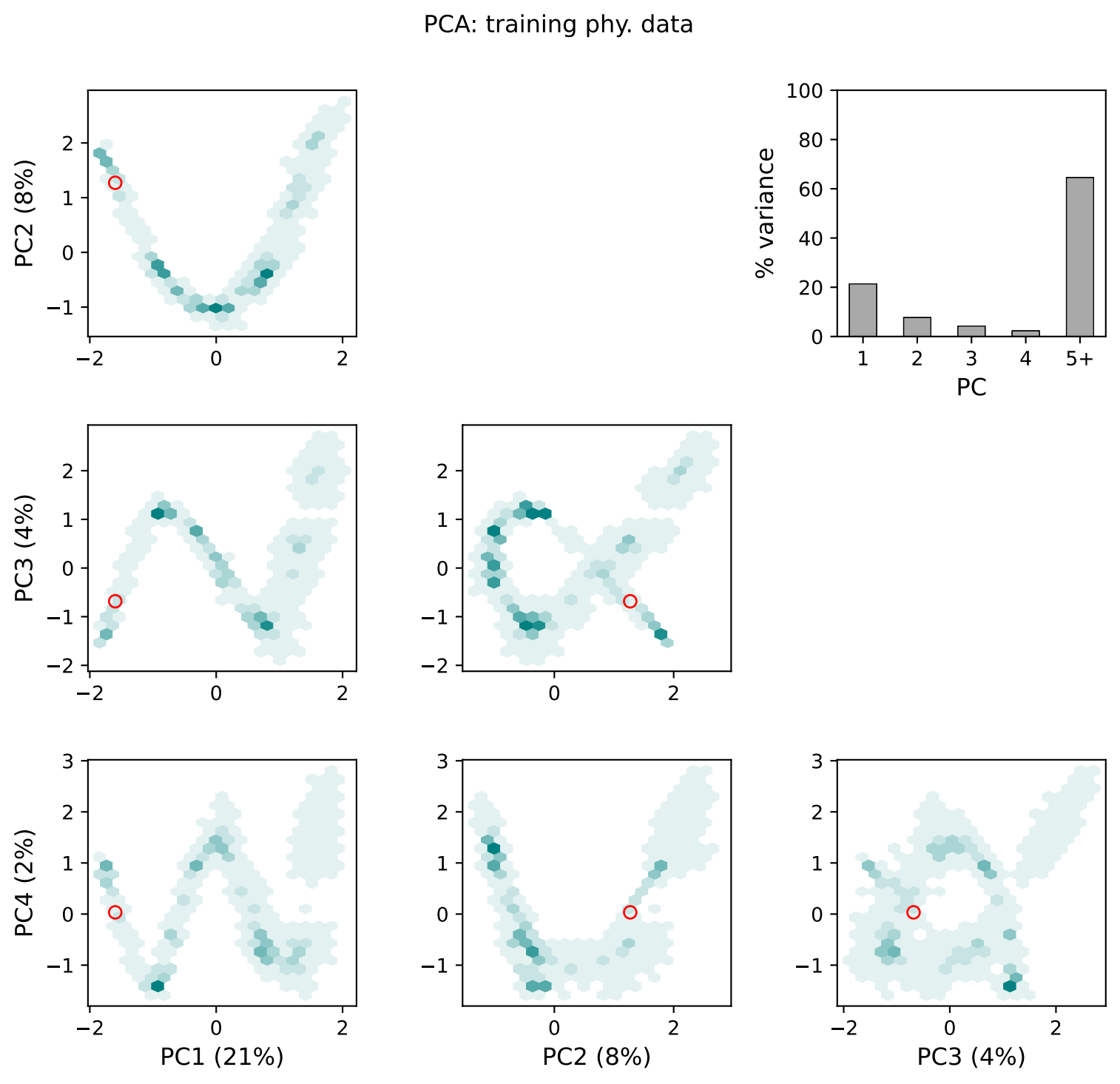
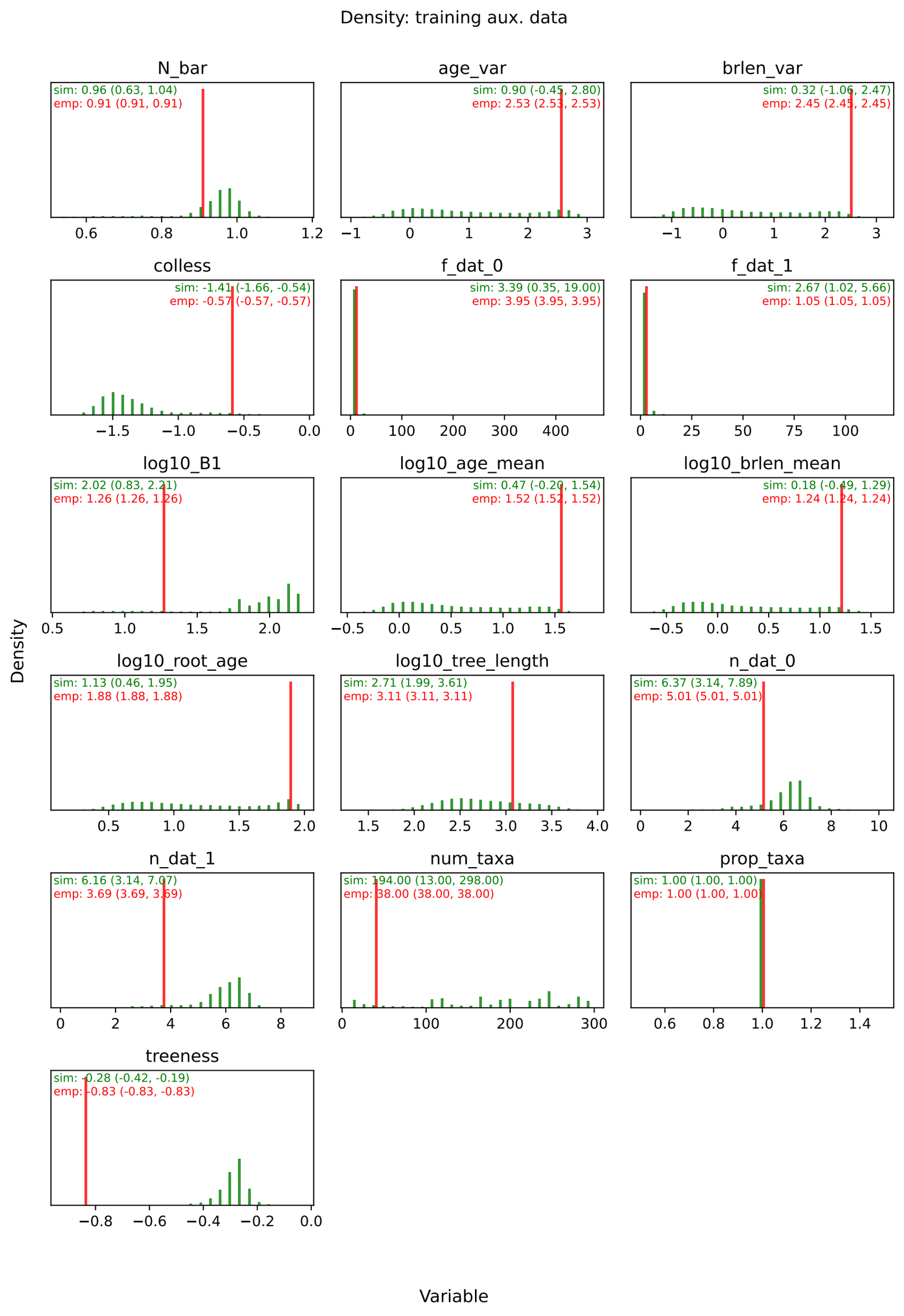
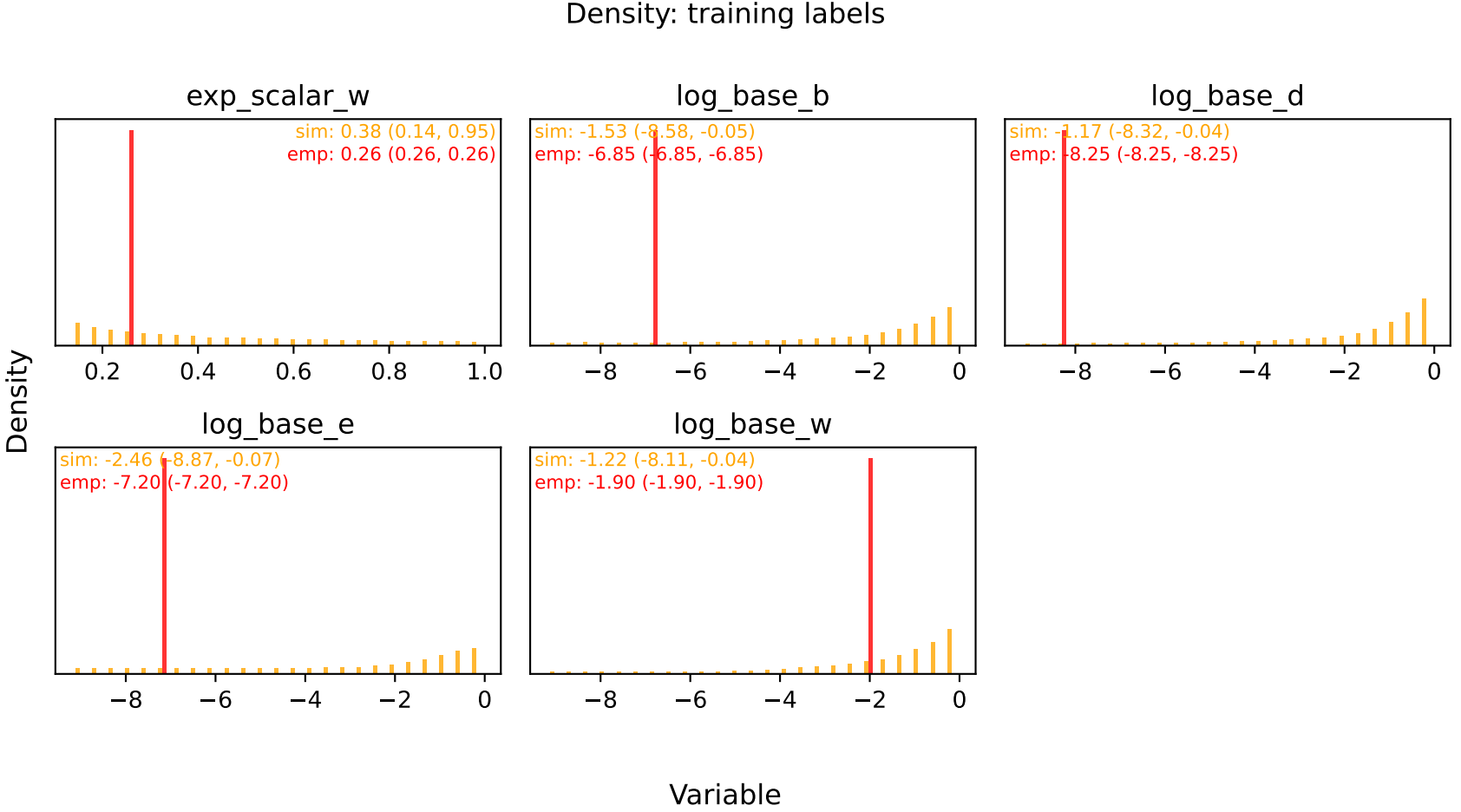

Estimate: sim.empirical.0
0
−2
−4
−6
−8
-1.17E+00
-1.90E+00
-2.63E+00
-5.96E+00
-7.20E+00
-8.60E+00
-6.71E+00
-8.25E+00
-8.70E+00
-1.53E+00
-6.85E+00
-8.94E+00
3.33E-01
1.26E-01
log_base_w
log_base_e
log_base_d
log_base_b
exp_scalar_w

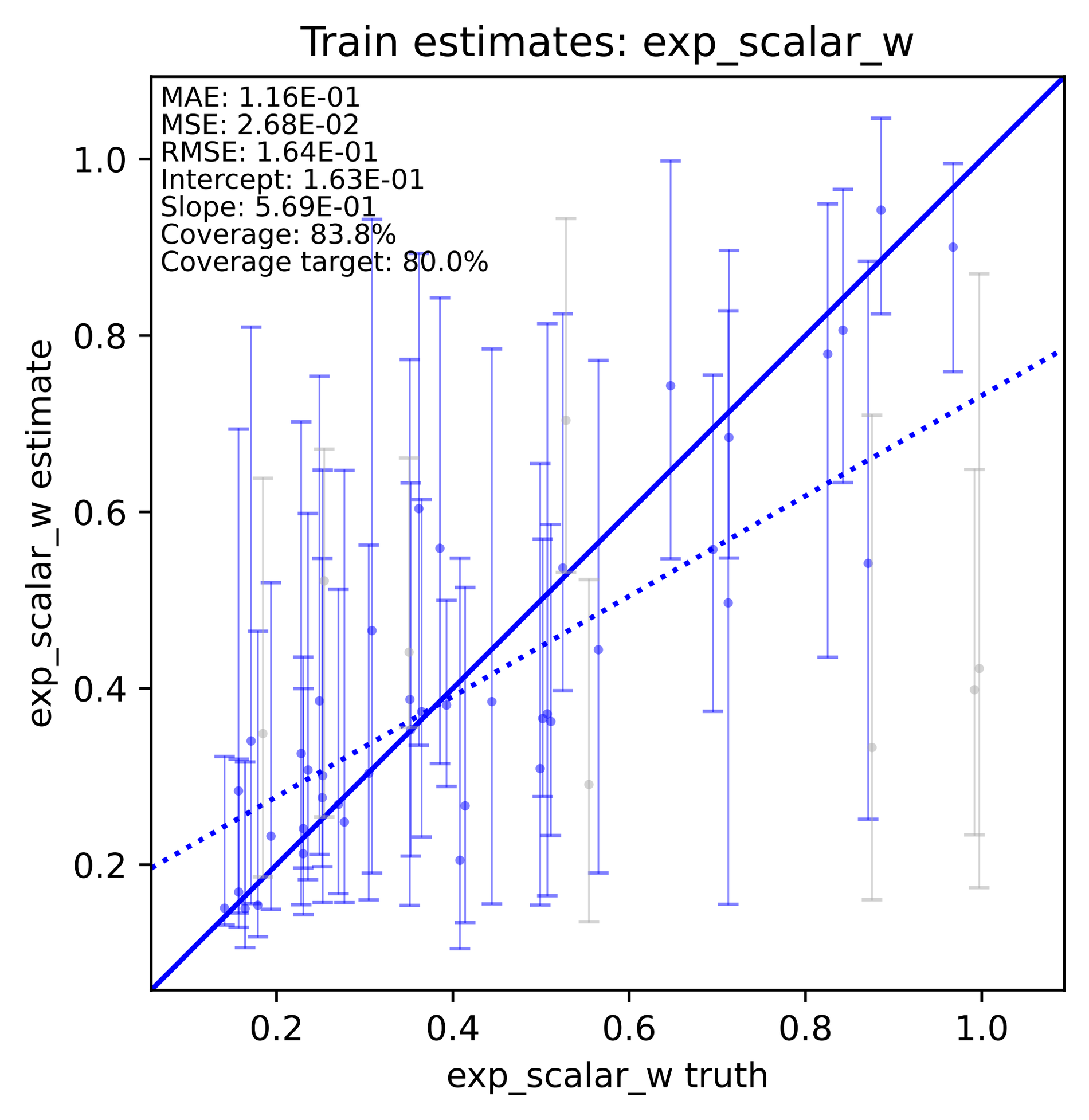

PCA: training aux. data
PC2 (20%)
% variance
PC
PC3 (15%)
PC4 (4%)
PC1 (54%)
PC2 (20%)
PC3 (15%)

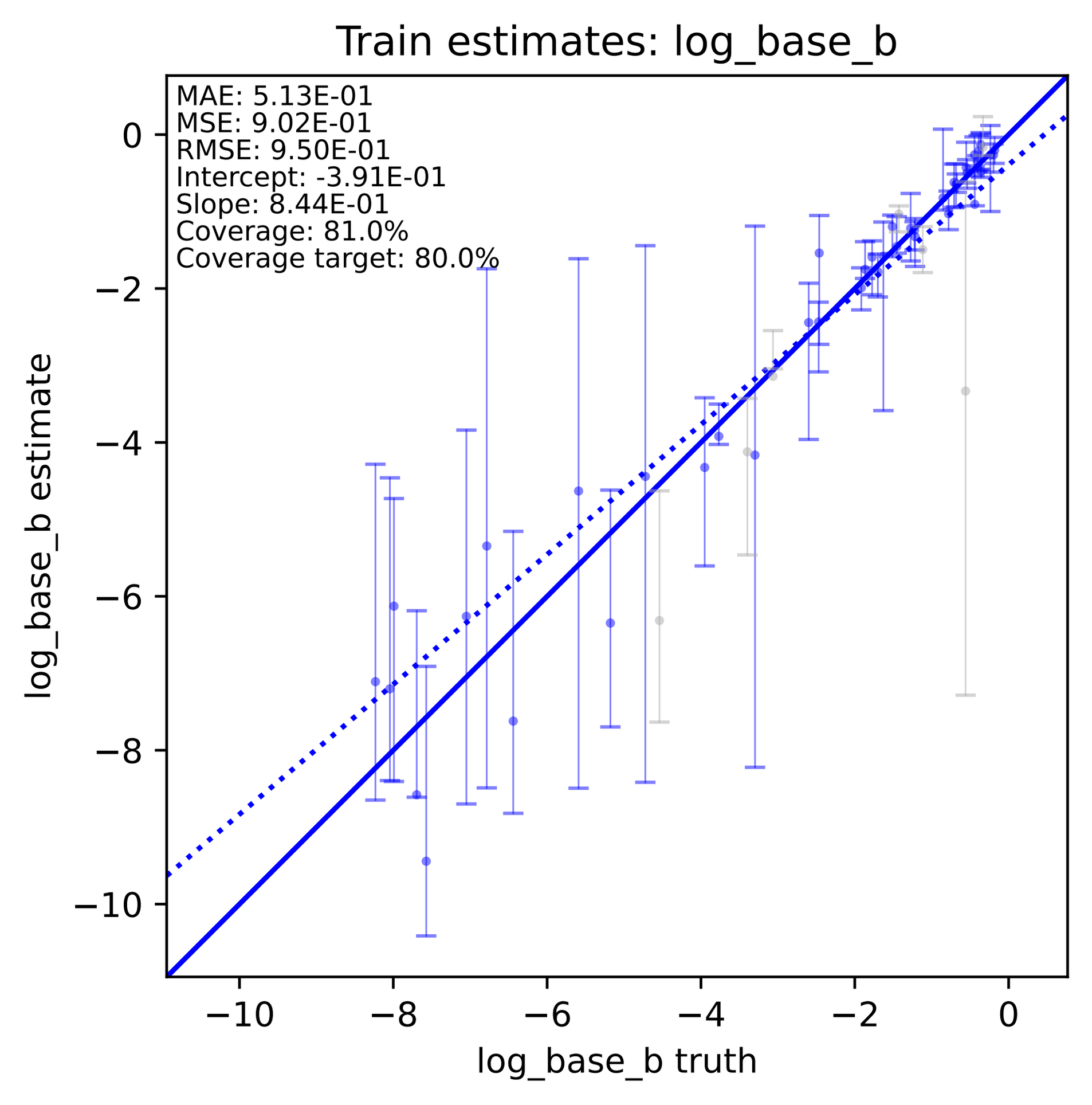

PCA: training labels
PC2 (20%)
% variance
PC
PC3 (20%)
PC4 (12%)
PC1 (42%)
PC2 (20%)
PC3 (20%)

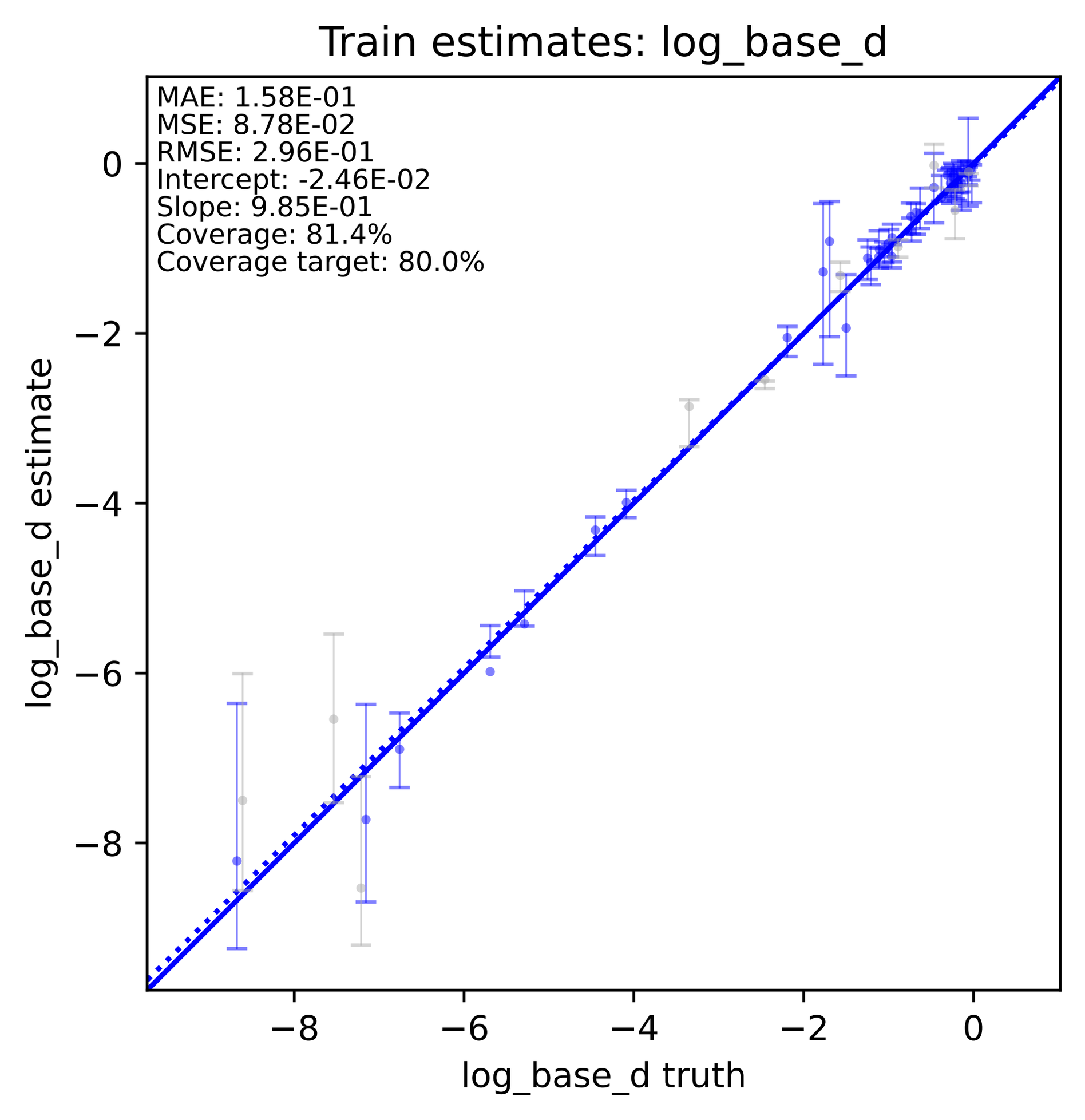

PCA: training phy. data
PC2 (8%)
% variance
PC
PC3 (4%)
PC4 (2%)
PC1 (21%)
PC2 (8%)
PC3 (4%)

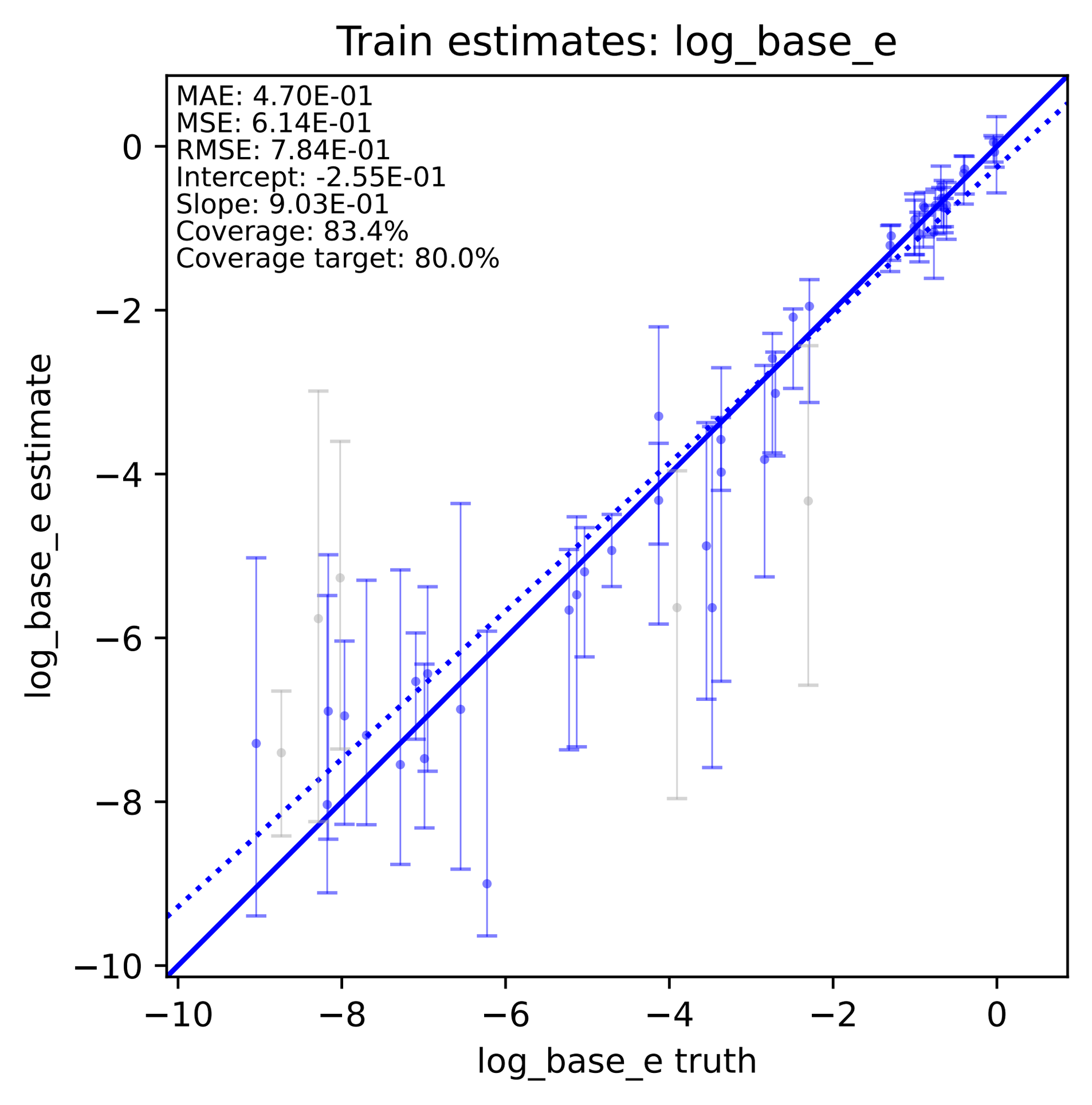

Density: training aux. data
N_bar
sim: 0.96 (0.63, 1.04)
emp: 0.91 (0.91, 0.91)
age_var
sim: 0.90 (-0.45, 2.80)
emp: 2.53 (2.53, 2.53)
brlen_var
sim: 0.32 (-1.06, 2.47)
emp: 2.45 (2.45, 2.45)
colless
sim: -1.41 (-1.66, -0.54)
emp: -0.57 (-0.57, -0.57)
f_dat_0
sim: 3.39 (0.35, 19.00)
emp: 3.95 (3.95, 3.95)
f_dat_1
sim: 2.67 (1.02, 5.66)
emp: 1.05 (1.05, 1.05)
log10_B1
sim: 2.02 (0.83, 2.21)
emp: 1.26 (1.26, 1.26)
log10_age_mean
sim: 0.47 (-0.20, 1.54)
emp: 1.52 (1.52, 1.52)
log10_brlen_mean
sim: 0.18 (-0.49, 1.29)
emp: 1.24 (1.24, 1.24)
log10_root_age
sim: 1.13 (0.46, 1.95)
emp: 1.88 (1.88, 1.88)
log10_tree_length
sim: 2.71 (1.99, 3.61)
emp: 3.11 (3.11, 3.11)
n_dat_0
sim: 6.37 (3.14, 7.89)
emp: 5.01 (5.01, 5.01)
n_dat_1
sim: 6.16 (3.14, 7.07)
emp: 3.69 (3.69, 3.69)
num_taxa
sim: 194.00 (13.00, 298.00)
emp: 38.00 (38.00, 38.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.28 (-0.42, -0.19)
emp: -0.83 (-0.83, -0.83)
Density
Variable

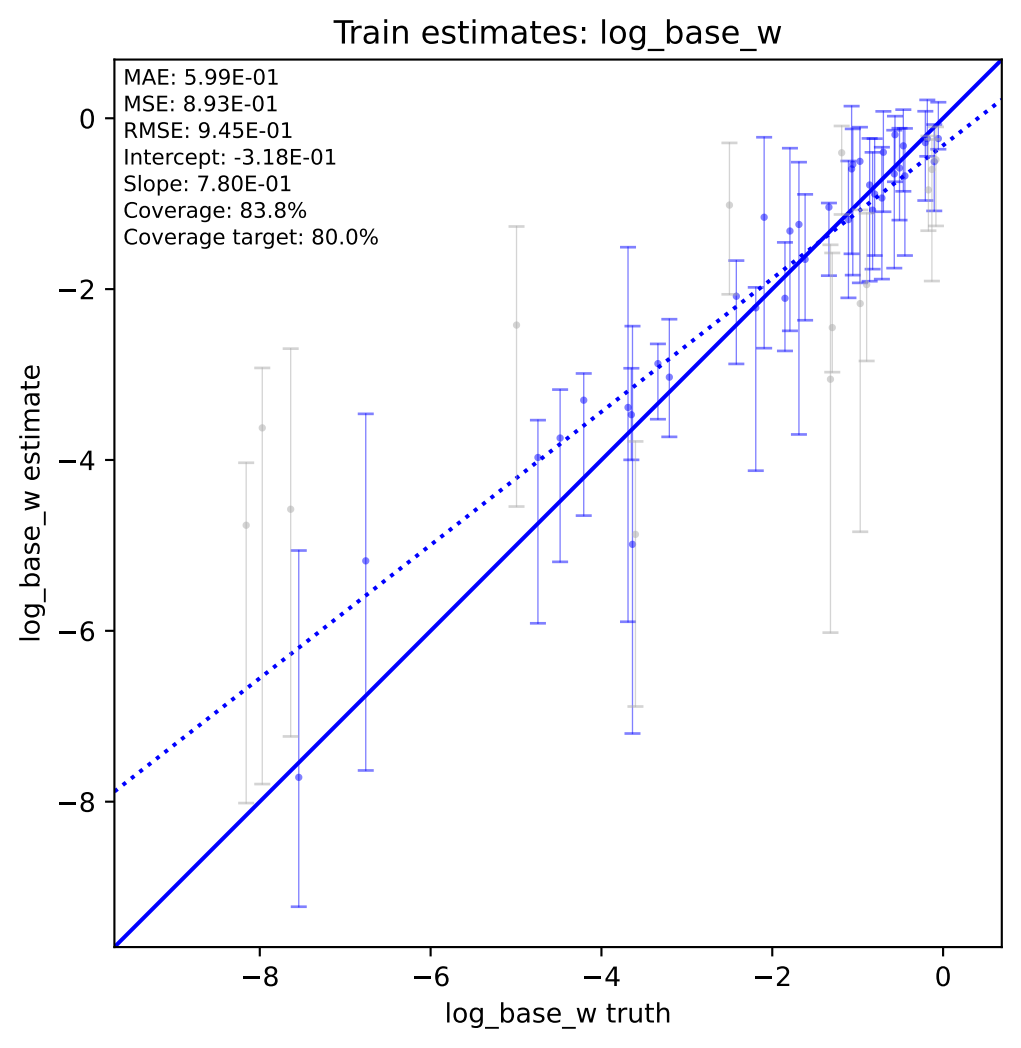

Density: training labels
exp_scalar_w
sim: 0.38 (0.14, 0.95)
emp: 0.26 (0.26, 0.26)
log_base_b
sim: -1.53 (-8.58, -0.05)
emp: -6.85 (-6.85, -6.85)
log_base_d
sim: -1.17 (-8.32, -0.04)
emp: -8.25 (-8.25, -8.25)
log_base_e
sim: -2.46 (-8.87, -0.07)
emp: -7.20 (-7.20, -7.20)
log_base_w
sim: -1.22 (-8.11, -0.04)
emp: -1.90 (-1.90, -1.90)
Density
Variable

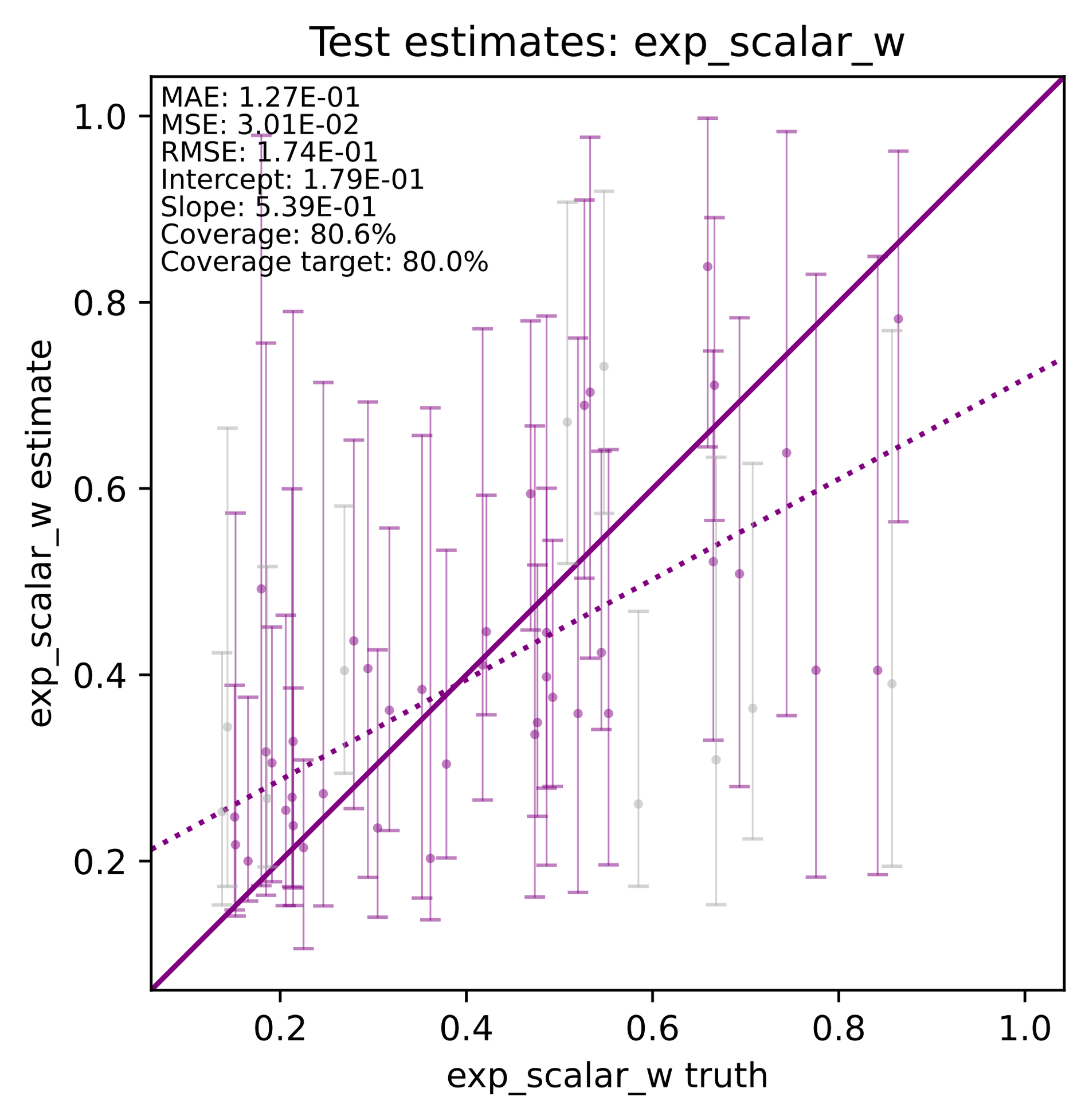

Train estimates: exp_scalar_w
MAE: 1.16E-01
MSE: 2.68E-02
RMSE: 1.64E-01
Intercept: 1.63E-01
Slope: 5.69E-01
Coverage: 83.8%
Coverage target: 80.0%
exp_scalar_w estimate
exp_scalar_w truth
0.2
0.4
0.6
0.8
1.0

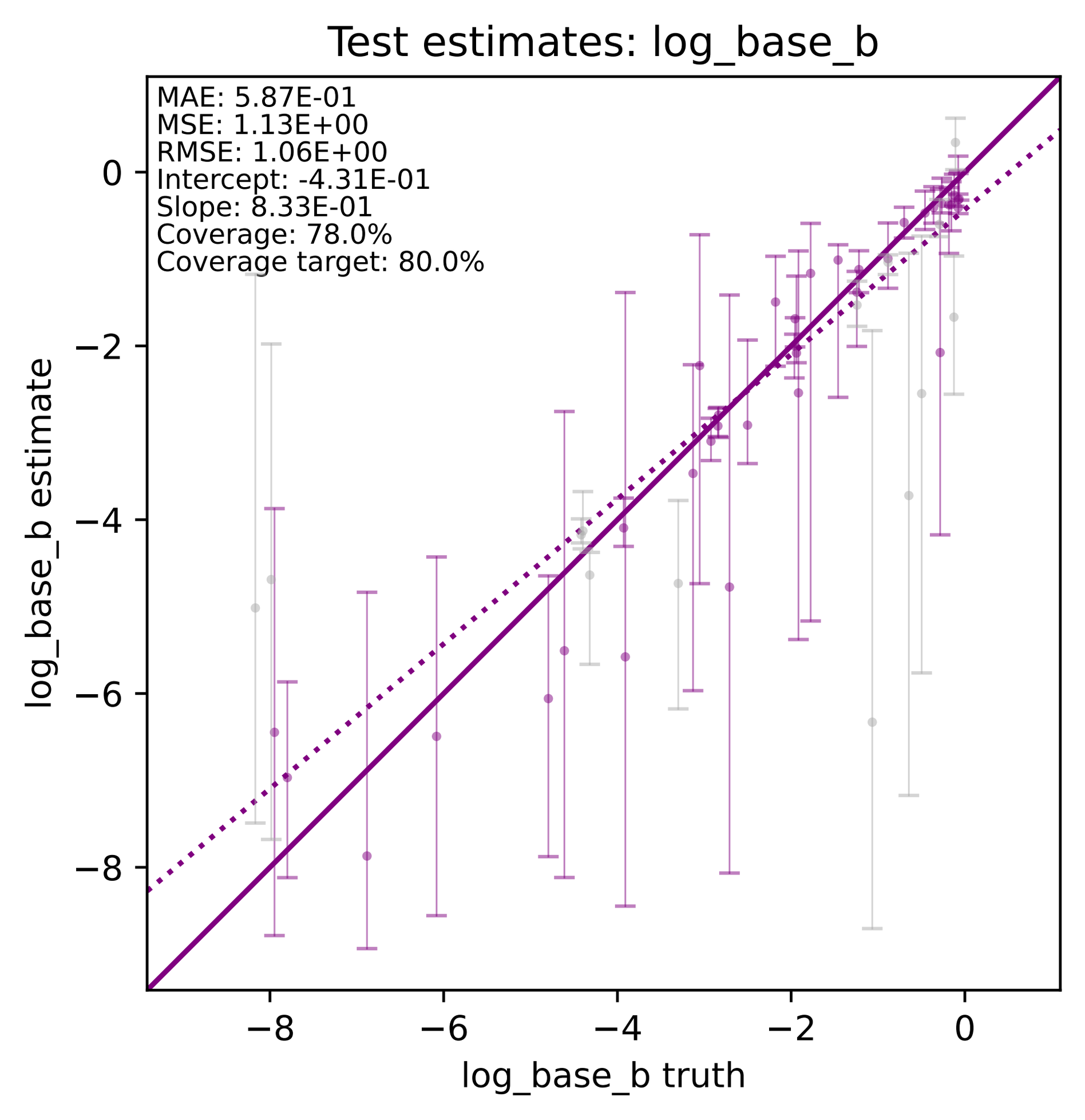

Train estimates: log_base_b
MAE: 5.13E-01
MSE: 9.02E-01
RMSE: 9.50E-01
Intercept: -3.91E-01
Slope: 8.44E-01
Coverage: 81.0%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

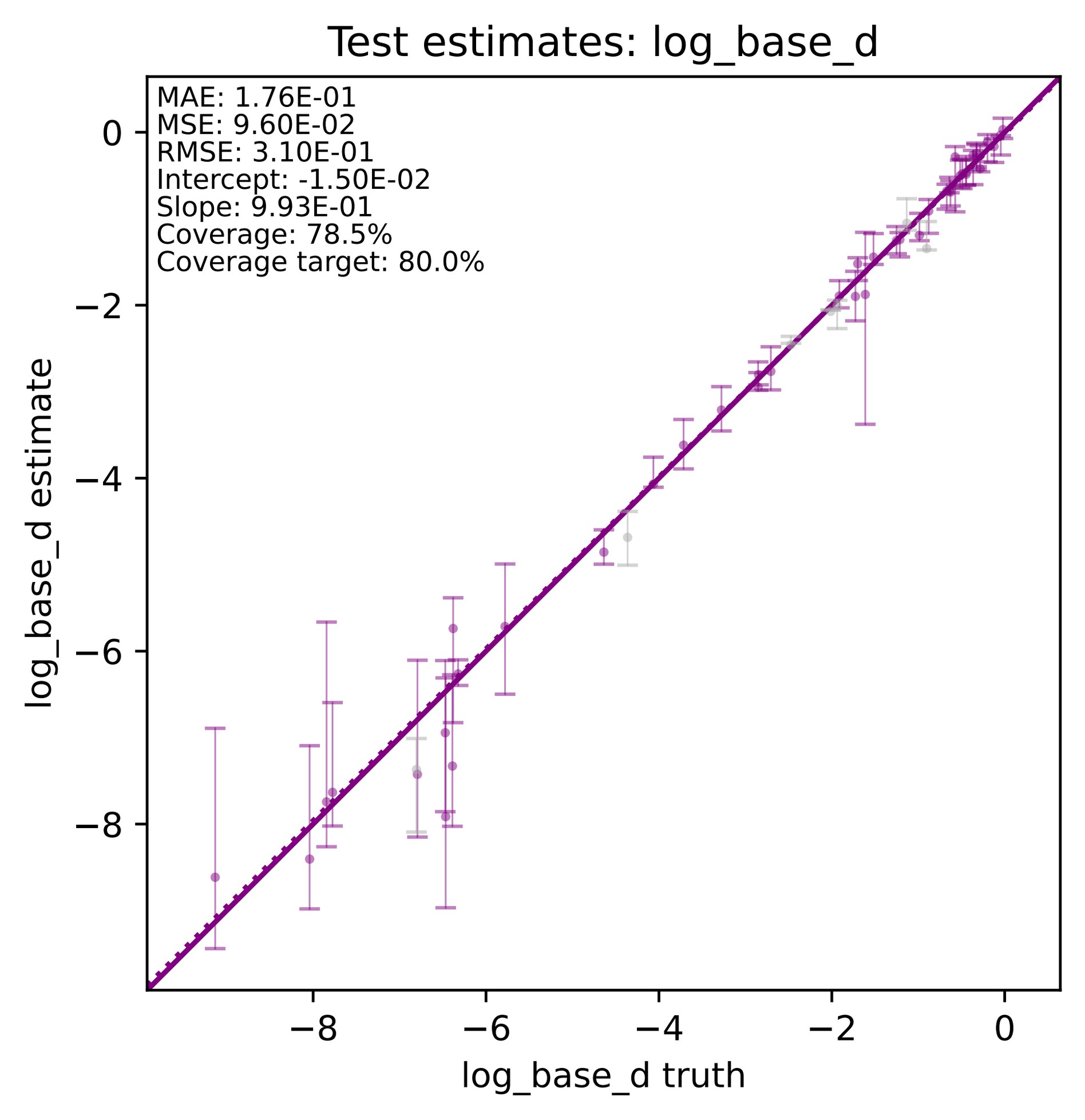

Train estimates: log_base_d
MAE: 1.58E-01
MSE: 8.78E-02
RMSE: 2.96E-01
Intercept: -2.46E-02
Slope: 9.85E-01
Coverage: 81.4%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

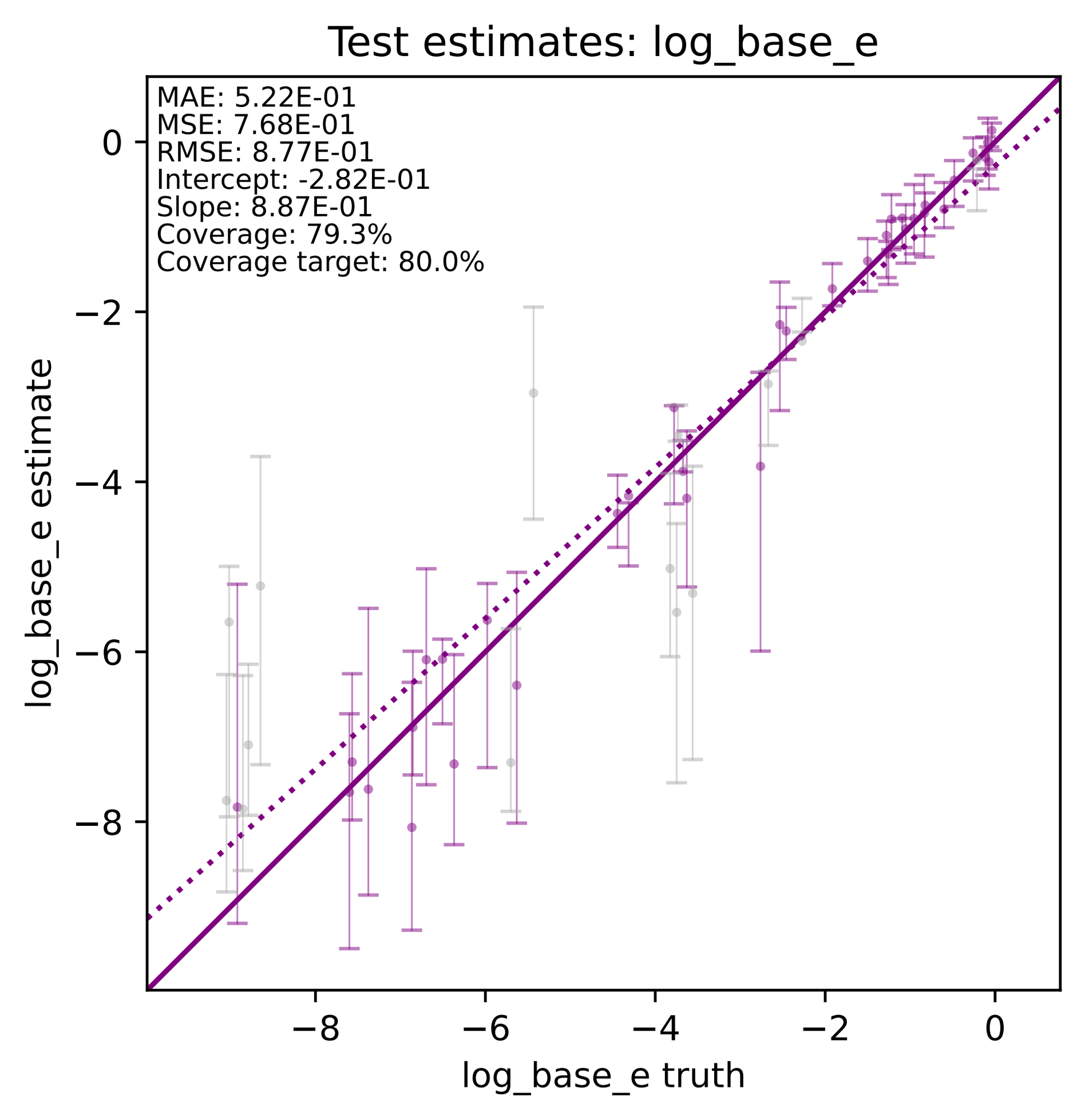

Train estimates: log_base_e
MAE: 4.70E-01
MSE: 6.14E-01
RMSE: 7.84E-01
Intercept: -2.55E-01
Slope: 9.03E-01
Coverage: 83.4%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

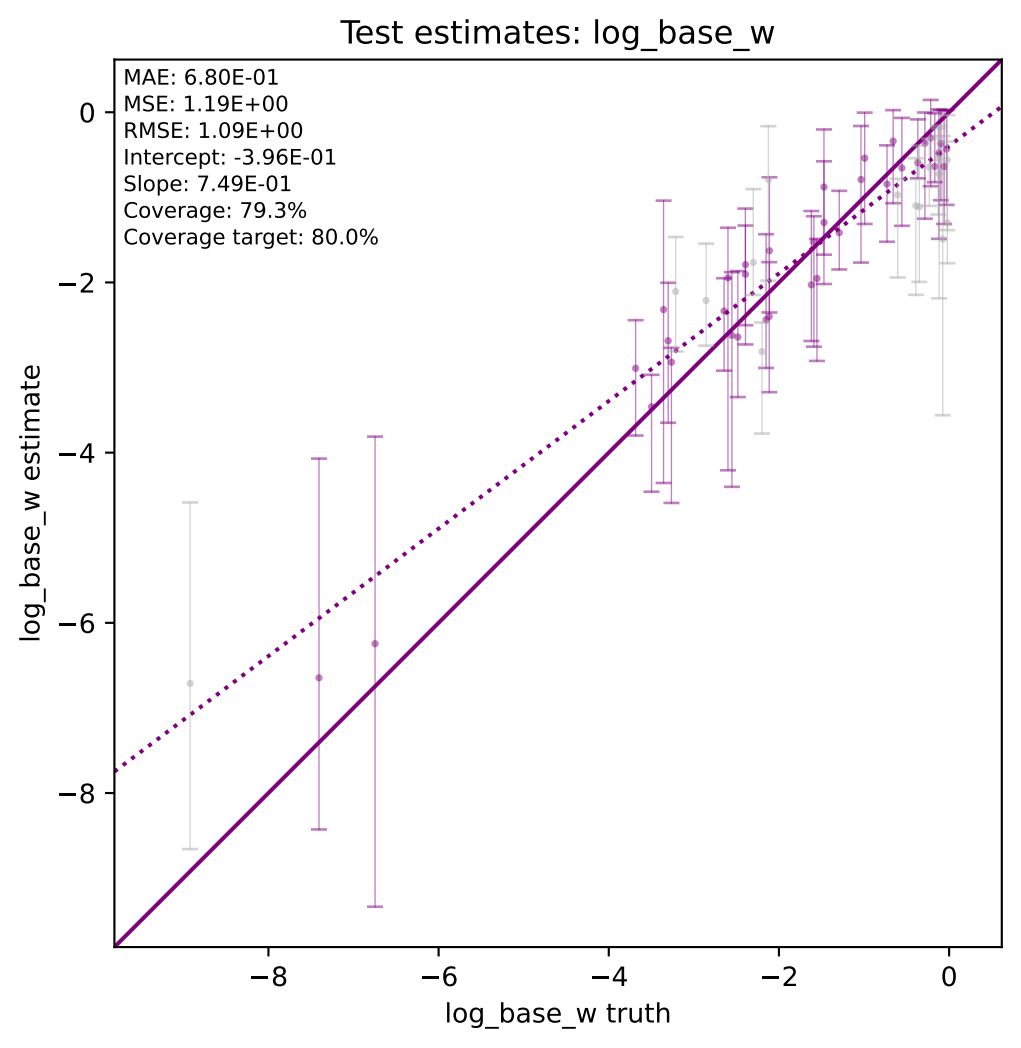

Train estimates: log_base_w
MAE: 5.99E-01
MSE: 8.93E-01
RMSE: 9.45E-01
Intercept: -3.18E-01
Slope: 7.80E-01
Coverage: 83.8%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

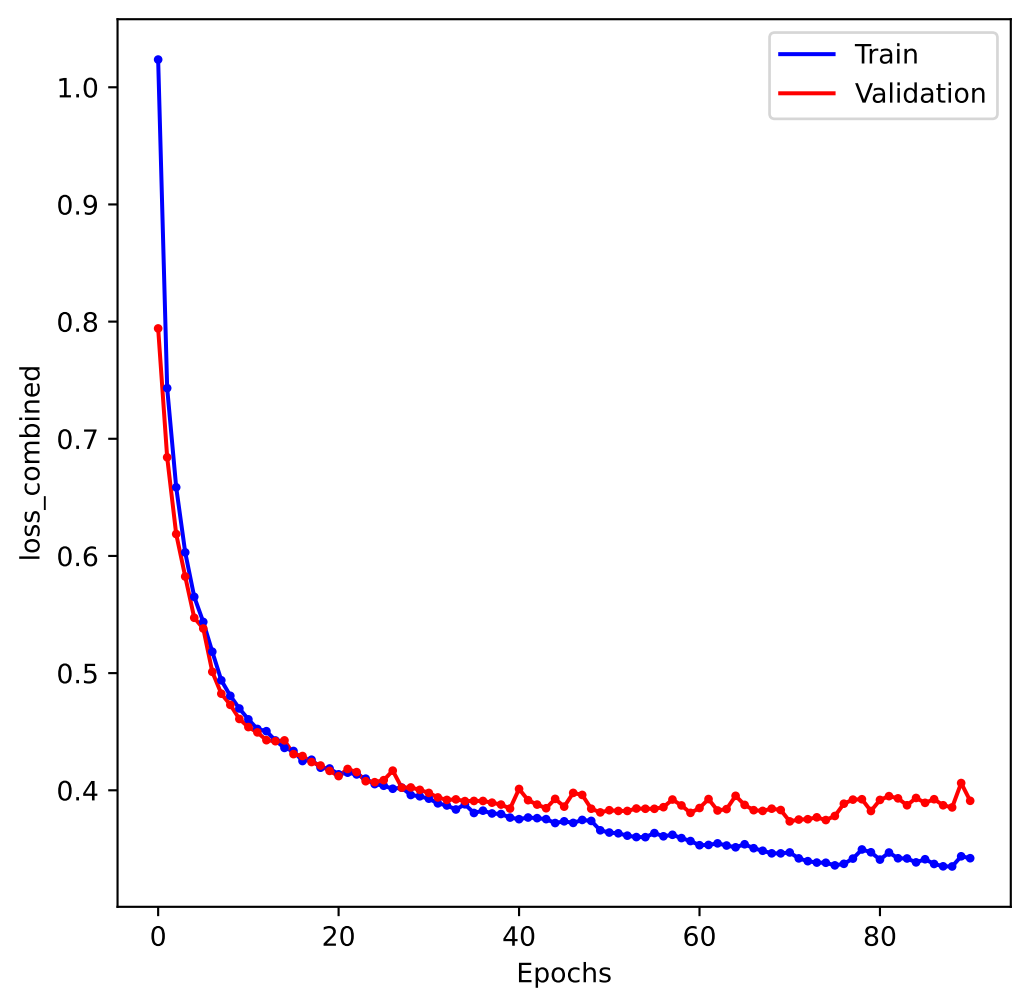

Test estimates: exp_scalar_w
MAE: 1.27E-01
MSE: 3.01E-02
RMSE: 1.74E-01
Intercept: 1.79E-01
Slope: 5.39E-01
Coverage: 80.6%
Coverage target: 80.0%
exp_scalar_w estimate
exp_scalar_w truth
0.2
0.4
0.6
0.8
1.0

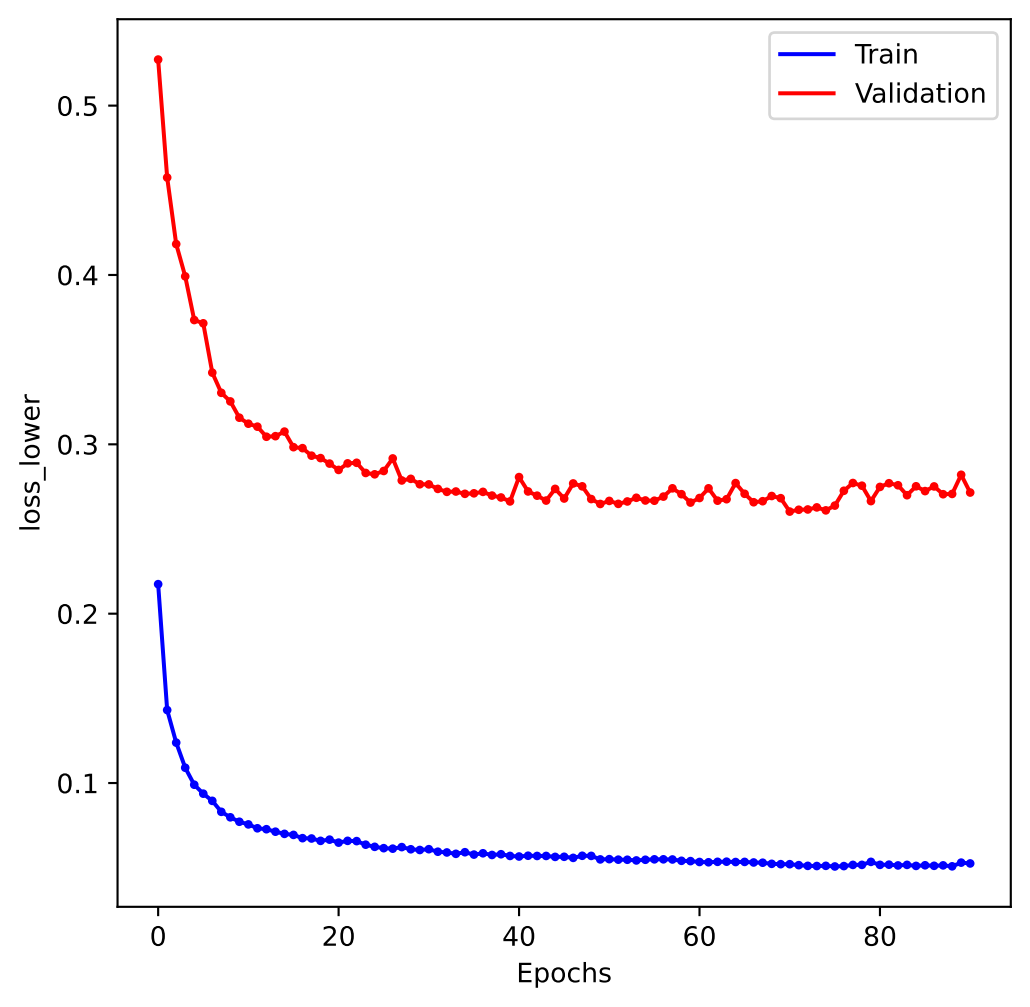

Test estimates: log_base_b
MAE: 5.87E-01
MSE: 1.13E+00
RMSE: 1.06E+00
Intercept: -4.31E-01
Slope: 8.33E-01
Coverage: 78.0%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

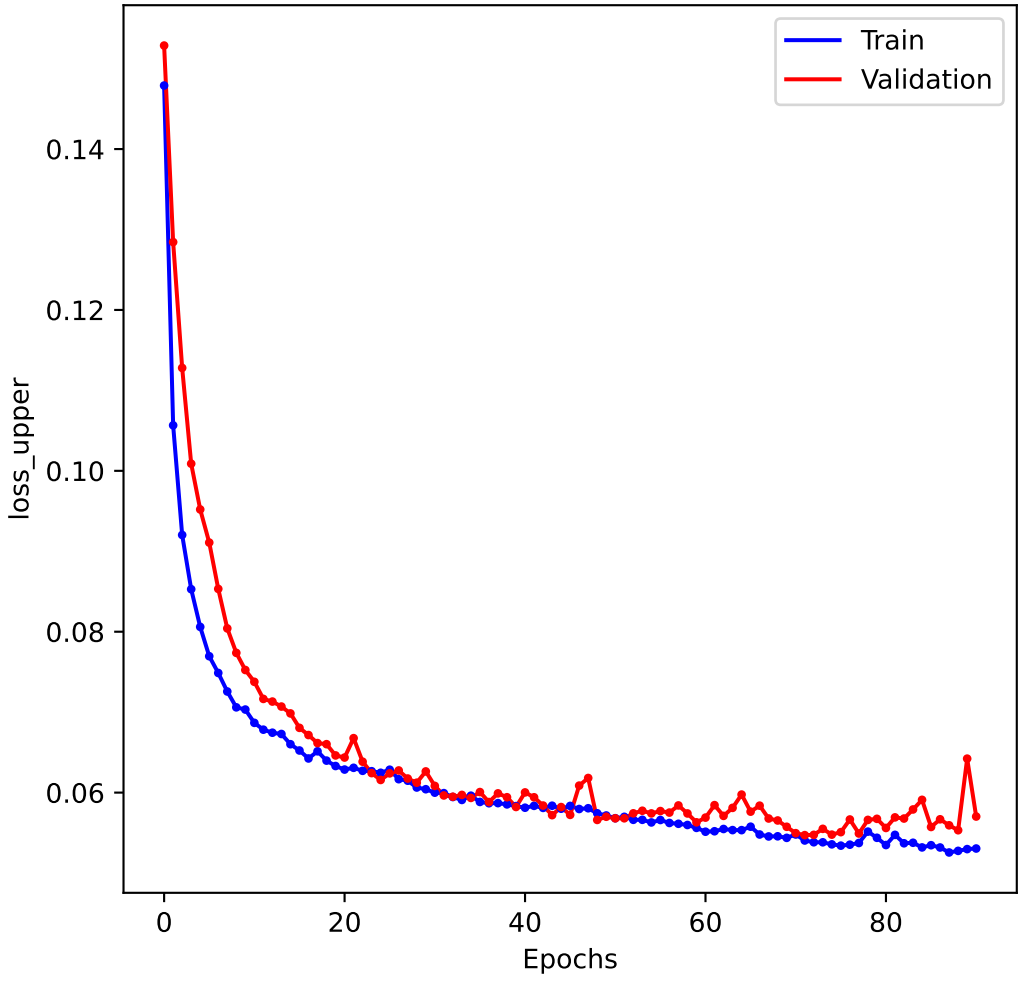

Test estimates: log_base_d
MAE: 1.76E-01
MSE: 9.60E-02
RMSE: 3.10E-01
Intercept: -1.50E-02
Slope: 9.93E-01
Coverage: 78.5%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

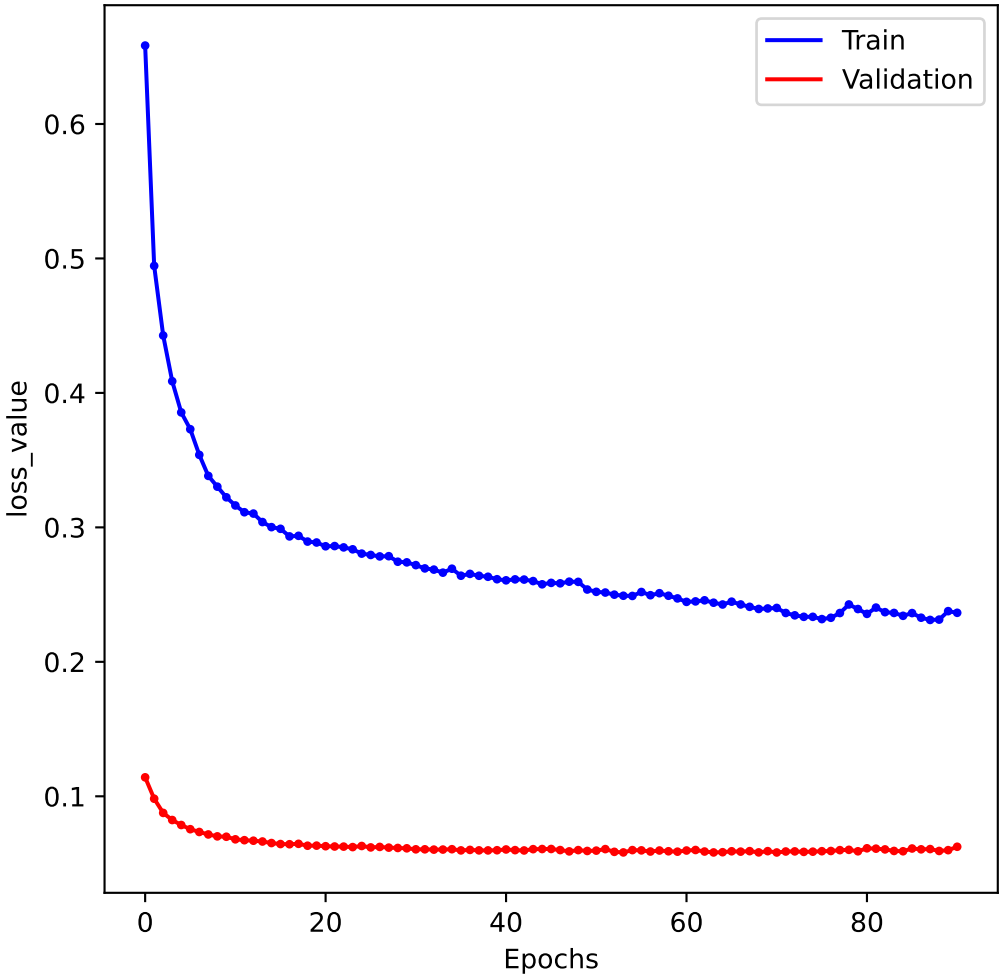

Test estimates: log_base_e
MAE: 5.22E-01
MSE: 7.68E-01
RMSE: 8.77E-01
Intercept: -2.82E-01
Slope: 8.87E-01
Coverage: 79.3%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

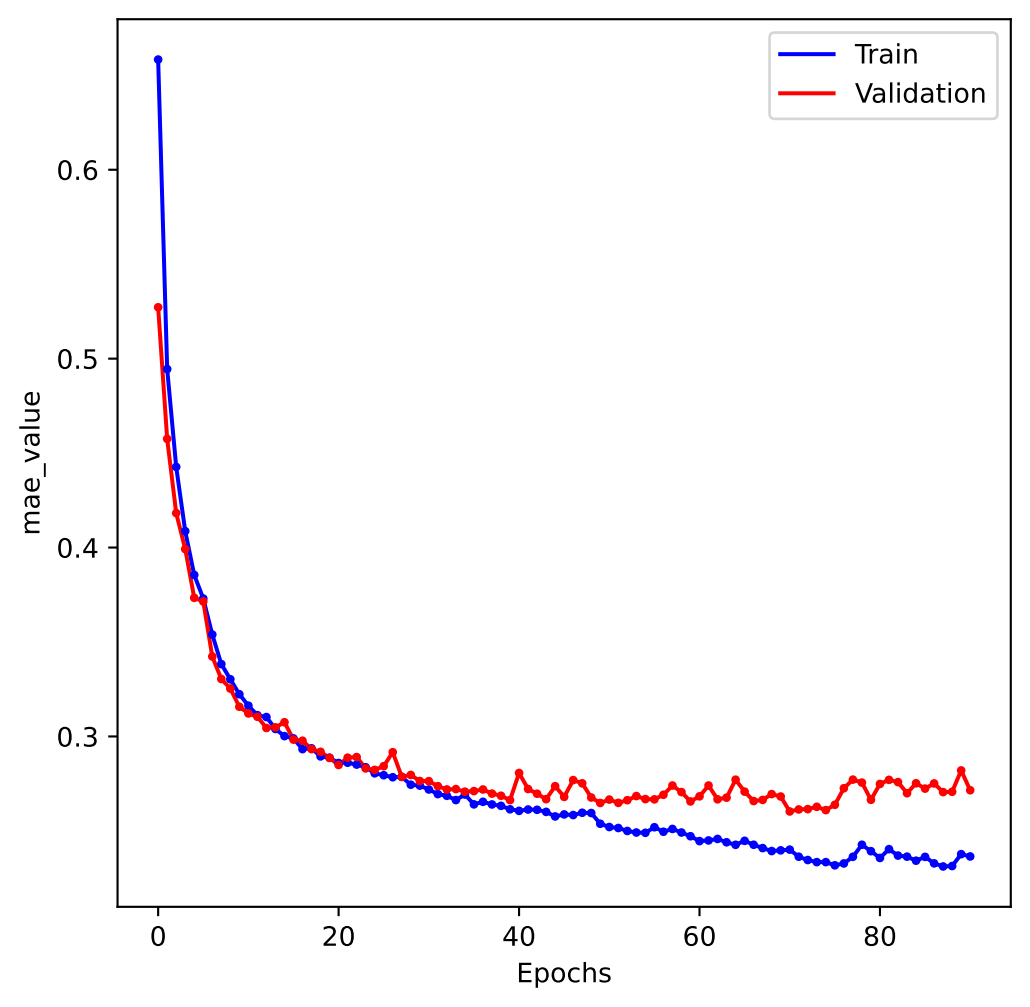

Test estimates: log_base_w
MAE: 6.80E-01
MSE: 1.19E+00
RMSE: 1.09E+00
Intercept: -3.96E-01
Slope: 7.49E-01
Coverage: 79.3%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

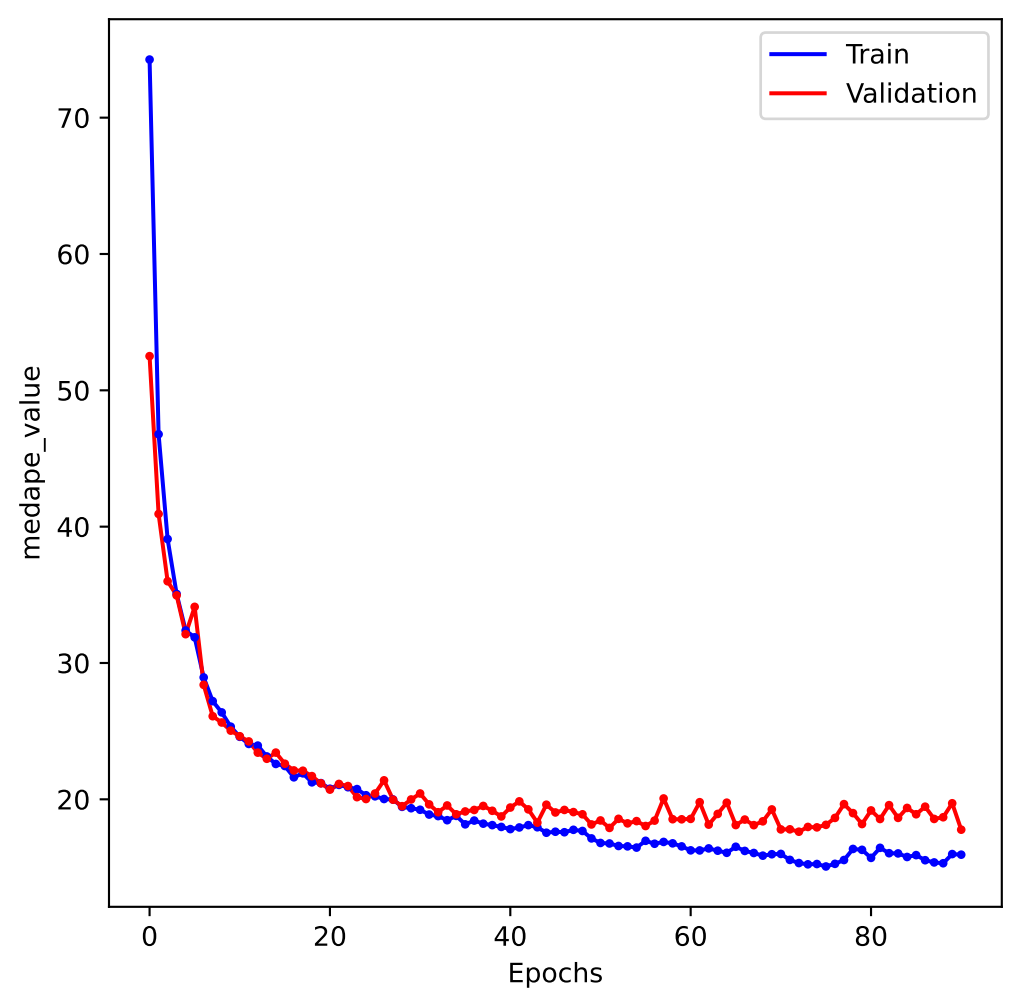

Train
Validation
loss_combined
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0
20
40
60
80
Epochs

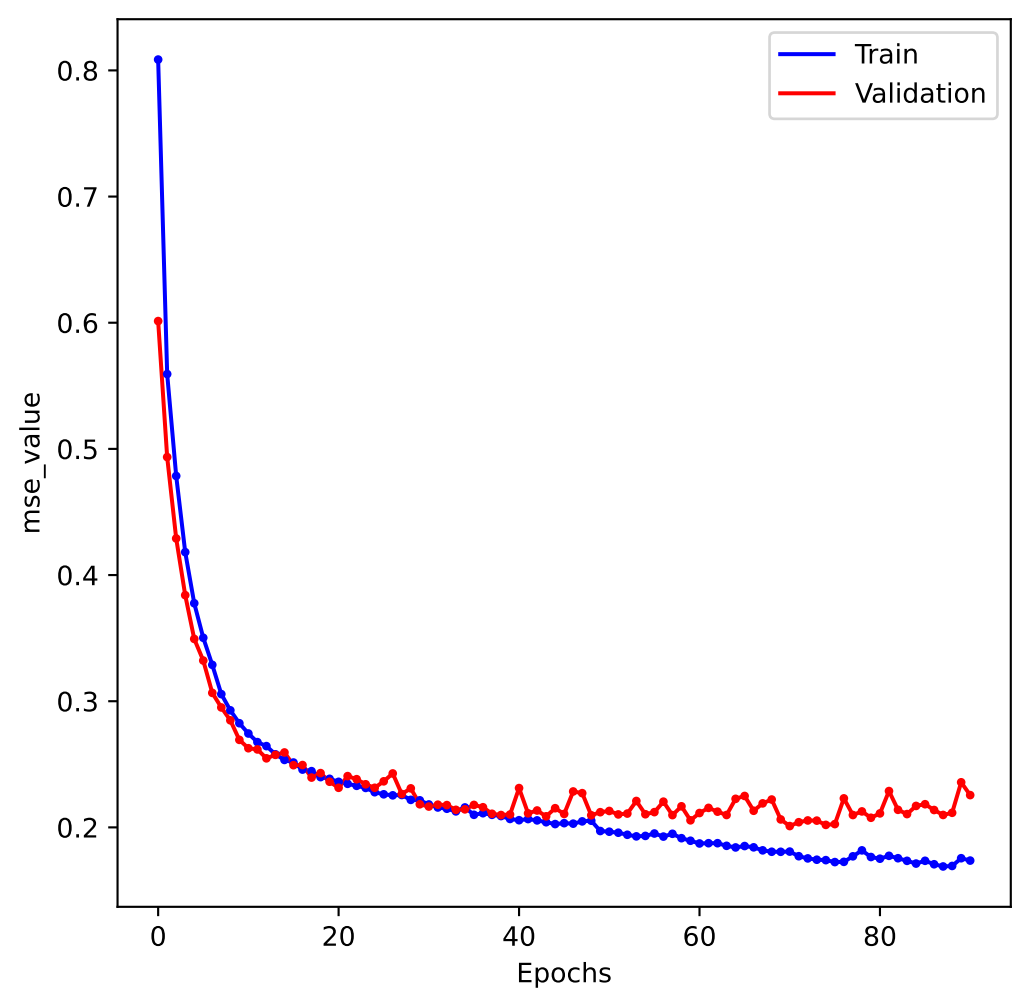

Train
Validation
loss_lower
0.5
0.4
0.3
0.2
0.1
0
20
40
60
80
Epochs

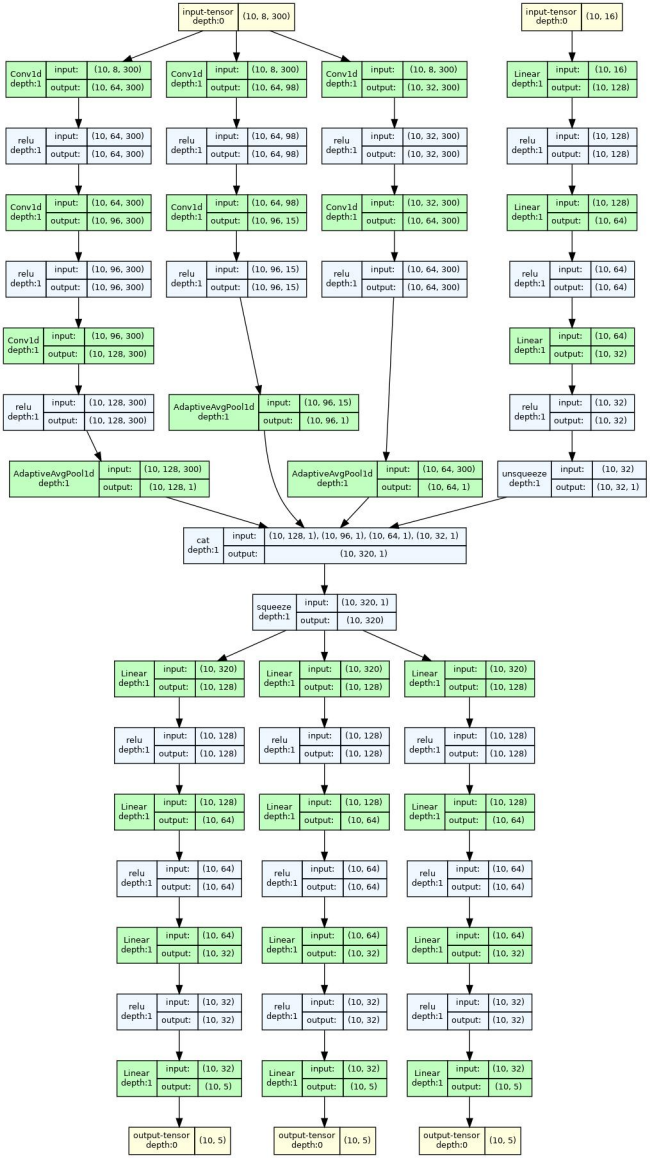

Train
Validation
loss_upper
0.14
0.12
0.10
0.08
0.06
0
20
40
60
80
Epochs


Train
Validation
loss_value
0.6
0.5
0.4
0.3
0.2
0.1
0
20
40
60
80
Epochs


Train
Validation
mae_value
0.6
0.5
0.4
0.3
0
20
40
60
80
Epochs


Train
Validation
70
60
50
40
30
20
medape_value
0
20
40
60
80
Epochs


Train
Validation
mse_value
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0
20
40
60
80
Epochs


input-tensor depth:0
(10, 8, 300)
input-tensor depth:0
(10, 16)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 300)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 98)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 32, 300)
Linear depth:1
input: (10, 16)
output: (10, 128)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64, 98)
output: (10, 64, 98)
relu depth:1
input: (10, 32, 300)
output: (10, 32, 300)
relu depth:1
input: (10, 128)
output: (10, 128)
Conv1d depth:1
input: (10, 64, 300)
output: (10, 96, 300)
Conv1d depth:1
input: (10, 64, 98)
output: (10, 96, 15)
Conv1d depth:1
input: (10, 32, 300)
output: (10, 64, 300)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 96, 300)
output: (10, 96, 300)
relu depth:1
input: (10, 96, 15)
output: (10, 96, 15)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64)
output: (10, 64)
Conv1d depth:1
input: (10, 96, 300)
output: (10, 128, 300)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 128, 300)
output: (10, 128, 300)
AdaptiveAvgPool1d depth:1
input: (10, 96, 15)
output: (10, 96, 1)
relu depth:1
input: (10, 32)
output: (10, 32)
AdaptiveAvgPool1d depth:1
input: (10, 128, 300)
output: (10, 128, 1)
AdaptiveAvgPool1d depth:1
input: (10, 64, 300)
output: (10, 64, 1)
unsqueeze depth:1
input: (10, 32)
output: (10, 32, 1)
cat depth:1
input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1)
output: (10, 320, 1)
squeeze depth:1
input: (10, 320, 1)
output: (10, 320)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
Linear depth:1
input: (10, 32)
output: (10, 5)
Linear depth:1
input: (10, 32)
output: (10, 5)
Linear depth:1
input: (10, 32)
output: (10, 5)
output-tensor depth:0
(10, 5)
output-tensor depth:0
(10, 5)
output-tensor depth:0
(10, 5)